

Hierarchical conditional empirical Bayes matrix factorization

Learning $p(L_k \mid L_{<k})$ for partially paired multi-omics

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Abstract

We study matrix factorizations whose loading distributions depend on earlier latent programs and fixed covariates. The target is a normalized directed prior, $p_\theta(L) = \prod_k p_{\theta_k}(L_k \mid L_{<k})$, learned jointly with the latent loadings. We derive a variational coordinate that combines the original cEBMF Gaussian likelihood calculation with uncertainty in parents and feedback from children. Profiling out the coordinate posterior gives a prior-learning objective that reduces to the original cEBNM marginal-likelihood problem when latent-covariate edges are absent. Ordinary ash updates are retained for feature factors. For partially paired ATAC–RNA data, two observation models are linked through their loading prior; completely unobserved terminal branches are integrated out to avoid an unnecessary variational penalty. A Gaussian-mixture implementation uses one-dimensional quadrature and fixed parent-integration points. The dependent prior is distinguished from the factorized posterior approximation and from the stronger alternative of correlated row blocks. Comparisons with TwinEB, learned VAE priors and multi-omics methods delimit the structural contribution without claiming unique representational capacity, causal identification or established predictive superiority.

Positioning. EBMF learns priors and factors through normal-means problems [1]; cEBMF allows fixed covariates to inform these priors [2]. Our target extends the covariates to uncertain loading coordinates while preserving that learning structure. The contribution to evaluate is the usefulness of explicit conditional program distributions and their compatible inference, not the general ability to represent dependence. Population mixtures, autoregressive priors and nonlinear multi-omics models already represent many forms of dependence.

Scope. The principal implementation uses Gaussian observations, conditional Gaussian mixtures (with optional exact zero spikes) for loadings, and ash on features. HMM feature smoothing remains an optional, separate prior. Exponential loading slabs and correlated posterior blocks are extensions, not part of the principal validated example. The supplied simulation is an executable illustration; a comparative scientific benchmark remains needed.

1 Conditional loading model

Let $Y \in \mathbb{R}^{n \times p}$, $L \in \mathbb{R}^{n \times K}$, and $F \in \mathbb{R}^{p \times K}$. Here $L_k = L_{:k}$ means the k th loading column, and $L_{<k} = L_{:,1:(k-1)}$. The observation model is

$$Y_{ij} \mid L, F, \tau \sim \mathcal{N}\left(\sum_{k=1}^K L_{ik} F_{jk}, \tau_{ij}^{-1}\right), \quad (i, j) \in \Omega. \quad (1)$$

The desired autoregressive loading prior is

$$p_\theta(L \mid X) = \prod_{k=1}^K p_{\theta_k}(L_k \mid L_{<k}, X). \quad (2)$$

As in the existing conditional-prior networks, we use the rowwise form

$$p_{\theta_k}(L_k \mid L_{<k}, X) = \prod_{i=1}^n g_{\theta_k}(L_{ik} \mid X_i, L_{i,<k}). \quad (3)$$

The parameters θ_k are shared across rows. If fixed covariates are unavailable, omit X_i . Conditional independence across rows in (3) is a specified architecture; it is not independence across loading columns.

For example, a CGB network learns

$$g_{\theta_k}(dv \mid P) = \pi_{k0}(P)\delta_0(dv) + [1 - \pi_{k0}(P)]\mathcal{N}(dv; \mu_k, s_k^2), \quad P = (X_i, L_{i,<k}).$$

Earlier loadings predict whether a later loading is active. An EMDN can also let earlier loadings predict component means and variances. Multiplying these normalized, acyclic conditionals defines a normalized joint prior. The ordering permits dependencies; it does not uniquely identify a tree.

2 Prior dependence and posterior approximation

The earlier loadings are latent, not observed covariates. A parent loading therefore receives information both from the matrix likelihood and from later loading-prior terms. For two loading coordinates in one row,

$$p(l_1, l_2) = g_1(l_1)g_2(l_2 | l_1) \implies p(l_1 | l_2, Y, F) \propto p(Y | l_1, l_2, F)g_1(l_1)g_2(l_2 | l_1).$$

This posterior feedback follows from the requested joint model itself. It does not introduce a reverse-direction generative prior.

2.1 Mixture labels and the variational family

For clarity, represent a loading mixture by a label Z_{ik} :

$$\begin{aligned} p(Z_{ik} = 0, L_{ik} \in dv | P_{ik}) &= \pi_{k0}(P_{ik})\delta_0(dv), \\ p(Z_{ik} = h, L_{ik} \in dv | P_{ik}) &= \pi_{kh}(P_{ik})\varphi(v; \mu_{kh}(P_{ik}), s_{kh}^2(P_{ik}))dv, \quad h > 0. \end{aligned} \quad (4)$$

Here P_{ik} contains the fixed covariates and the selected earlier parents. These labels only identify existing mixture components. They are not an additional representation or upper latent layer.

Use the variational approximation

$$q(L, Z, F) = \prod_{i,k} q_{ik}(L_{ik}, Z_{ik}) \prod_{j,k} q_{jk}^F(F_{jk}). \quad (5)$$

Each value stays dependent on its own label. The generative prior (2) remains dependent even though this posterior approximation factorizes across loading coordinates. Thus this method *learns the requested conditional prior*, but does not represent posterior covariance between distinct loading coordinates. Section 10 states the alternative if that covariance is also required.

For overlapping Gaussian components, augmenting the labels before making the mean-field approximation can give a different approximation from marginalizing them first. Both describe the same generative marginal model. With fixed parents and no children, the local optimum is the usual normal-means posterior, so augmentation does not spoil the cEBMF limit.

2.2 A single objective

Let $g_{\eta_k}^F$ be the ordinary ash prior on feature column k , with mixture weights estimated during fitting. The unpenalized ELBO is

$$\begin{aligned} \mathcal{L}(q, \theta, \eta, \tau) &= \mathbb{E}_q \log p(Y | L, F, \tau) + H(q) \\ &\quad + \sum_{i,k} \mathbb{E}_q \log g_{\theta_k}(L_{ik}, Z_{ik} | P_{ik}) + \sum_{j,k} \mathbb{E}_q \log g_{\eta_k}^F(F_{jk}). \end{aligned} \quad (6)$$

For the loading terms, densities and entropy are taken with respect to a label-specific reference measure: unit mass at zero for the spike and Lebesgue measure for continuous components. Gaussian densities are not added to spike probability mass at zero.

3 Deriving a loading-coordinate update

3.1 The original cEBMF likelihood calculation

Write $w_{ij} = M_{ij}\tau_{ij}$, where M is the observation mask. Let bars denote first posterior moments, and define

$$\begin{aligned}\bar{R}_{ij,-k} &= Y_{ij} - \sum_{h \neq k} \bar{L}_{ih} \bar{F}_{jh}, \\ a_{ik} &= \sum_j w_{ij} \mathbb{E}_q[F_{jk}^2], & b_{ik} &= \sum_j w_{ij} \bar{R}_{ij,-k} \bar{F}_{jk}.\end{aligned}\tag{7}$$

All sums involving observed Y are over observed entries. Expanding the expected squared error while holding the other variational factors fixed gives the loading-dependent likelihood term

$$\mathbb{E}_{q_{-ik}} \log p(Y \mid L, F, \tau) = -\frac{1}{2}a_{ik}v^2 + b_{ik}v + \text{constant}, \quad v = L_{ik}.\tag{8}$$

The denominator uses $\mathbb{E}[F_{jk}^2]$, not $(\mathbb{E}F_{jk})^2$, as in ordinary cEBMF. F need not be sampled.

3.2 The own-prior and child terms

Define

$$A_{ik}(v, z; \theta_k) = \mathbb{E}_{q(P_{ik})} \log g_{\theta_k}(v, z \mid P_{ik}).\tag{9}$$

Only latent parents are averaged; the external covariates are fixed. Changing v also changes every direct child's prior. Their contribution is

$$D_{ik}(v) = \sum_{c \in \text{ch}(k)} \mathbb{E}_{q(L_{ic}, Z_{ic})q(P_{ic} \setminus L_{ik})} \log g_{\theta_c}(L_{ic}, Z_{ic} \mid P_{ic}[v]).\tag{10}$$

Under the total ordering, every later coordinate is a possible direct child; structurally omitted network inputs create no edge. Holding the other q factors and parameters fixed, the variational coordinate is

$$q_{ik}^*(v, z) \propto \exp\{-a_{ik}v^2/2 + b_{ik}v + A_{ik}(v, z; \theta_k) + D_{ik}(v)\}.\tag{11}$$

This follows by maximizing the terms of (6) involving q_{ik} : expected log joint plus its entropy.

Two necessary corrections to plug-in conditioning. Average the *log prior* over uncertain parents and include the children's prior terms. In general, $\mathbb{E} \log g(v \mid P) \neq \log g(v \mid \mathbb{E}P)$ and $\mathbb{E} \log g(v \mid P) \neq \log \mathbb{E}g(v \mid P)$. Using fitted earlier loadings as fixed inputs does not perform these steps.

4 Learning the conditional prior and posterior together

Write the exponent in (11) as $\Psi_{ik}(v, z; \theta_k)$, and define its normalizer

$$\mathcal{Z}_{ik}(\theta_k) = \sum_z \int \exp\{\Psi_{ik}(v, z; \theta_k)\} d\nu_z(v). \quad (12)$$

The network parameters are distinct for different loading columns. Consequently D is fixed when optimizing θ_k , even though it varies with the candidate loading v . The Gibbs variational identity gives

$$\max_{q_{ik}} \{\mathbb{E}_{q_{ik}} \Psi_{ik} + H(q_{ik})\} = \log \mathcal{Z}_{ik}(\theta_k). \quad (13)$$

The joint prior-and-posterior update is therefore

$$\boxed{\theta_k^{\text{new}} \in \arg \max_{\theta_k} \sum_i \log \mathcal{Z}_{ik}(\theta_k), \quad q_{ik}^{\text{new}} = q_{ik}^*(\cdot; \theta_k^{\text{new}}).} \quad (14)$$

The neural prior is learned from all rows using the same ELBO as the loading inference. An exact block optimizer cannot decrease that ELBO. Approximate integration and finite neural optimization require their own accuracy controls; they do not inherit an automatic monotonicity guarantee.

4.1 Proof of the reduction to ordinary cEBMF

With no latent-covariate edges, there are neither uncertain parents nor child messages:

$$A_{ik}(v, z; \theta_k) = \log g_{\theta_k}(v, z \mid X_i), \quad D_{ik}(v) = 0.$$

For $a = a_{ik} > 0$, set $x = b_{ik}/a$ and $s^2 = 1/a$. Then

$$e^{-av^2/2+bv} = \sqrt{2\pi/a} e^{b^2/(2a)} \varphi(x; v, 1/a).$$

After summing over the component label,

$$\begin{aligned} \log \mathcal{Z}_{ik}(\theta_k) &= \log \left\{ \int \varphi(x; v, s^2) g_{\theta_k}(dv \mid X_i) \right\} \\ &\quad + \frac{1}{2} \log(2\pi/a) + b^2/(2a). \end{aligned} \quad (15)$$

The last two terms do not depend on θ_k . Thus (14) is **exactly the original cEBNM marginal-likelihood fit**, and (11) gives its posterior. Feature and noise updates below also reduce to their ordinary cEBMF forms.

In code, use this analytic limit and call the existing cEBNM solver with the same options, initialization and update schedule. A single complete-data EM M-step, even using infinitely many exact posterior draws, is not generally the same update as this marginal-likelihood fit.

For $a = b = 0$ use (11) directly. Also, do not renormalize $\exp(A)$ as an ordinary prior and discard its normalizer: with uncertain parents that normalizer generally depends on θ_k .

5 Gaussian mixtures and exact zero spikes

Suppress the row and loading indices. For each continuous component h , the expected log own prior is quadratic in v . Define

$$\begin{aligned}\alpha_h &= a + \mathbb{E}_P[s_h(P)^{-2}], \\ \beta_h &= b + \mathbb{E}_P[\mu_h(P)s_h(P)^{-2}], \\ \gamma_h &= \mathbb{E}_P \log \pi_h(P) - \frac{1}{2} \mathbb{E}_P \log(2\pi s_h(P)^2) - \frac{1}{2} \mathbb{E}_P[\mu_h(P)^2 s_h(P)^{-2}].\end{aligned}\quad (16)$$

Assume these expectations are finite and $\alpha_h > 0$. Put $m_h = \beta_h/\alpha_h$, $V_h = 1/\alpha_h$, and

$$I_{rh} = \int v^r \varphi(v; m_h, V_h) e^{D(v)} dv, \quad r = 0, 1, 2. \quad (17)$$

Completing the square gives unnormalized component masses

$$\begin{aligned}W_h &= \exp\{\gamma_h + \beta_h^2/(2\alpha_h)\} \sqrt{2\pi/\alpha_h} I_{0h}, \quad h > 0, \\ W_0 &= \exp\{\mathbb{E}_P \log \pi_0(P) + D(0)\}.\end{aligned}\quad (18)$$

The zero-spike likelihood factor is one in the natural-parameter form $e^{-av^2/2+bv}$. There is no omitted Gaussian evidence constant in W_0 . Normalize with $r_h = W_h/\sum_t W_t$. Then

$$\begin{aligned}q(v \mid z = h) &= \frac{\varphi(v; m_h, V_h) e^{D(v)}}{I_{0h}}, \quad h > 0, \\ \mathbb{E}_q v &= \sum_{h>0} r_h I_{1h}/I_{0h}, & \mathbb{E}_q v^2 &= \sum_{h>0} r_h I_{2h}/I_{0h}.\end{aligned}\quad (19)$$

Omit the spike for an unspiked EMDN. Compute component masses in log space. With deterministic parents and $D = 0$ these are the ordinary analytic Gaussian-mixture normal-means formulas.

5.1 How to calculate the child message

For child c , let $r_{ch} = q(Z_c = h)$, $u_{ch} = \mathbb{E}(L_c \mid Z_c = h)$, and $T_{ch} = \mathbb{E}(L_c^2 \mid Z_c = h)$. A Gaussian component contributes

$$J_{ch}(P) = -\frac{1}{2} \log(2\pi s_{ch}^2(P)) - \frac{T_{ch} - 2\mu_{ch}(P)u_{ch} + \mu_{ch}(P)^2}{2s_{ch}^2(P)}. \quad (20)$$

Consequently,

$$D(v) = \sum_{c \in \text{ch}(k)} \sum_h r_{ch} \mathbb{E}_{P_c \setminus v} [\log \pi_{ch}(P_c[v]) + J_{ch}(P_c[v])], \quad (21)$$

with $J_{c0} = 0$ for the fixed zero atom. For CGB with a global slab, J is constant in v and drops from the v -dependent tilt. For EMDN, its means and variances depend on parents, so J must remain. Child labels and values can be integrated using their probabilities and moments rather than sampled.

6 A two-loading example of the desired dependency

As a transparent illustration, suppose the loadings are strictly binary:

$$L_1 \sim \text{Bernoulli}(\rho), \quad L_2 \mid L_1 \sim \text{Bernoulli}\{s(\alpha + \beta L_1)\}, \quad s(t) = \text{sigmoid}(t). \quad (22)$$

This directly models $p(L_2 \mid L_1)$. Under $q(L_1)q(L_2)$, let $r_1 = q(L_1 = 1)$ and $r_2 = q(L_2 = 1)$. Since $v^2 = v$ for binary v , the likelihood log-odds contribution is $b_k - a_k/2$. The child update is

$$\text{logit}(r_2^{\text{new}}) = b_2 - a_2/2 + \alpha + \beta r_1. \quad (23)$$

For the parent, write $s_0 = s(\alpha)$ and $s_1 = s(\alpha + \beta)$. Its update contains feedback from the inferred child activation:

$$\begin{aligned} \text{logit}(r_1^{\text{new}}) &= b_1 - a_1/2 + \text{logit}(\rho) \\ &\quad + r_2 \log(s_1/s_0) + (1 - r_2) \log\{(1 - s_1)/(1 - s_0)\}. \end{aligned} \quad (24)$$

If $\beta > 0$, a confidently active child tends to favor parent activation. If $\beta = 0$, the child correction vanishes and both equations reduce to ordinary independent Bernoulli-prior normal-means updates.

This example distinguishes three objects: the learned conditional prior in (22), the posterior approximation r_1, r_2 , and the posterior feedback required by that prior. A factorized q does not remove the conditional relationship being learned in p.

7 Keeping the feature update as ordinary ash

Under the scalar-factorized approximation (5), set

$$\begin{aligned} a_{jk}^F &= \sum_i w_{ij} \mathbb{E}_q[L_{ik}^2], \\ b_{jk}^F &= \sum_i w_{ij} \left(Y_{ij} - \sum_{h \neq k} \bar{L}_{ih} \bar{F}_{jh} \right) \bar{L}_{ik}. \end{aligned} \quad (25)$$

Fit the existing ash normal-means model at b_{jk}^F/a_{jk}^F , with SE $(a_{jk}^F)^{-1/2}$. Update its mixture weights and posterior moments at each fitting iteration. Its configured penalty and other options are preserved.

This is the original feature-side empirical Bayes update. Merely drawing F from a prior fitted during initialization would not implement this learning algorithm. No HMM or feature-side neural model is needed for this derivation.

8 Noise, entropy and regularization

8.1 Observation noise

Under (5), the expected squared residual is

$$\begin{aligned}\mathbb{E}_q[(Y_{ij} - L_i^T F_j)^2] &= (Y_{ij} - \bar{L}_i^T \bar{F}_j)^2 \\ &+ \sum_k \{\mathbb{E}[L_{ik}^2] \mathbb{E}[F_{jk}^2] - \bar{L}_{ik} \bar{F}_{jk}\}.\end{aligned}\quad (26)$$

For unknown constant precision,

$$\tau^{\text{new}} = \frac{|\Omega|}{\sum_{ij \in \Omega} \mathbb{E}_q[(Y_{ij} - L_i^T F_j)^2]}.\quad (27)$$

The corresponding row- or column-specific versions use the same sums restricted to their groups. Known noise remains fixed.

8.2 Entropy of the actual variational approximation

For a continuous loading component in (19),

$$H_h = \frac{1}{2} \log(2\pi V_h) + \frac{\mathbb{E}_h[(v - m_h)^2]}{2V_h} - \mathbb{E}_h[D(v)] + \log I_{0h}.\quad (28)$$

The value-and-label entropy is $-\sum_h r_h \log r_h + \sum_{h>0} r_h H_h$. The atom has zero within-component entropy. For F, use the ordinary EBNM identity to compute its prior KL contribution. Thus the ELBO can be evaluated for the actual q used by the algorithm, rather than a separate distribution fitted afterward to sampler marginals.

When later coordinates or priors change, a previously updated q must stay fixed until its next explicit update. A stored tilted density cannot silently reevaluate itself using newly changed parent distributions or neural parameters. Its representation must preserve the state that defined it.

8.3 A consistent sparsity penalty

First validate the unpenalized case. The existing neural spike regularizer can then be included as

$$R(q, \theta) = \sum_{i,k} (\lambda_k - 1) \mathbb{E}_q \log \pi_{k0}(P_{ik}), \quad \lambda_k \geq 1.\quad (29)$$

The own penalty is constant in the candidate L_{ik} , so it can be added outside the log normalizer during θ_k optimization. A child's penalty depends on L_{ik} and must be included in $D(v)$. Without latent parents these become the fixed-covariate parameter penalties, preserving the ordinary penalized cEBNM limit.

This is a regularized ELBO, or a bound for an augmented observation model; it is not the unpenalized model's evidence. The ash global mixture-weight penalty has its existing, separate semantics and should be preserved.

9 The learning algorithm

One cEBMF iteration. Retain the original residual-based L_k/F_k schedule. Extend only the loading fit to account for uncertain parents and child feedback.

1. Initialize L, F and their moments using the existing cEBMF scheme. Preserve supplied factors and their scaling. There is no universal statistical justification for thresholding initial loadings at 0.5.
2. For $k = 1, \dots, K$:
 - (a) Calculate a^L, b^L from current variational moments using (7).
 - (b) If k has no latent parents or children, call its original cEBNM solver. Otherwise compute (9)–(10) and jointly fit the conditional prior and posterior through (14).
 - (c) Store loading component probabilities, moments and the fixed variational representation needed for later expectations and entropy.
 - (d) Update F_k with the original ash fit using (25), and refresh residuals.
3. Update unknown noise using the expected residuals and evaluate the same variational objective.
4. Repeat until the chosen iteration limit or convergence criterion.

The target remains $\prod_k p_\theta(L_k \mid L_{<k})$. This variational algorithm does not require a final full-matrix MCMC burn-in and sampling stage.

9.1 Where numerical approximation remains

Each integral over the target loading in (17) is one-dimensional. Adaptive quadrature or Gauss–Hermite integration around the Gaussian base components can evaluate it. Expectations over several uncertain parents can require separate quadrature or Monte Carlo integration. Keep parent integration nodes or draws fixed during a local parameter fit, and check results by increasing integration accuracy.

Sharp child constraints can make these calculations demanding. A practical implementation must assess runtime and approximation error; the derivation does not guarantee a fast neural optimizer or a predictive improvement. When there are no latent edges, however, use the analytic cEBNM route and avoid introducing numerical approximation unnecessarily.

9.2 Compatibility and convergence requirements

The no-edge route should preserve the old prior options, initialization, RNG behavior, L_k/F_k schedule, noise updates and pruning policy. With a hierarchy, deleting a factor changes child input identities and needs an explicit graph/network policy. Fixing rank is reasonable for an initial controlled comparison only when the comparator uses the same rank.

The exact variational argument gives nondecrease of the ELBO for exact block fits. Existing approximate scalar solvers and finite neural updates are not all exact maximizers; behavioral compatibility must not be confused with an unconditional monotonicity theorem.

10 A structured posterior alternative

The same conditional-prior model can also be fitted with posterior covariance between loading coordinates. For this additional inference choice, consider

$$q(L, F) = \prod_i q_i(L_i) \prod_{j,k} q_{jk}^F(F_{jk}).$$

The optimal row coordinate is

$$q_i^*(L_i) \propto p_\theta(L_i | X_i) \exp\{-\frac{1}{2}L_i^T H_i L_i + h_i^T L_i\}, \quad (30)$$

where

$$H_i = \sum_j w_{ij} \mathbb{E}[F_j F_j^T], \quad h_i = \sum_j w_{ij} Y_{ij} \mathbb{E}[F_j].$$

This retains the same conditional-prior model, but the feature numerator must now use cross moments:

$$b_{jk}^F = \sum_i w_{ij} \left[Y_{ij} \mathbb{E}[L_{ik}] - \sum_{h \neq k} \mathbb{E}[L_{ik} L_{ih}] \mathbb{E}[F_{jh}] \right]. \quad (31)$$

Relative to (25), the extra term is

$$- \sum_i w_{ij} \sum_{h \neq k} \text{Cov}(L_{ik}, L_{ih}) \bar{F}_{jh}.$$

Thus one cannot sample a correlated L block and pass only marginal means and variances into an unchanged mean-residual feature update.

Full row blocks do not automatically recover mean-field cEBMF when prior edges vanish: the likelihood can still correlate loading coordinates. For exact reduction, define blocks by connected components of the prior graph and split them into singleton columns in the no-edge case. Singleton blocks recover ordinary cEBNM; larger blocks require a multivariate profile objective.

11 Partially paired ATAC–RNA factorization

Construct the union of cell identifiers. Each modality has its own features, rank, noise and observation mask:

$$Y^A = L^A(F^A)^\top + E^A, \quad Y^R = L^R(F^R)^\top + E^R.$$

The joint loading prior is

$$\begin{aligned} p(L_i^A, L_i^R | X_i) &= \prod_{k=1}^{K_A} g_k^A(L_{ik}^A | X_i, L_{i,<k}^A) \\ &\times \prod_{k=1}^{K_R} g_k^R(L_{ik}^R | X_i, L_i^A, L_{i,<k}^R). \end{aligned} \quad (32)$$

This is the same autoregressive construction with all ATAC nodes ordered before RNA nodes. It permits different ranks and does not require a common loading matrix. An ash prior is learned separately for each feature column. The directed prior describes a conditional association, not an identified causal direction.

11.1 Observed likelihood and the three cell groups

Let $m_i^A, m_i^R \in \{0, 1\}$ indicate whether a modality has any observed entries in cell i . Writing each modality likelihood using only its observed features, the row likelihood at fixed feature factors is

$$\int p(L_i^A | X_i) p(L_i^R | L_i^A, X_i) p(Y_i^{A,\text{obs}} | L_i^A)^{m_i^A} p(Y_i^{R,\text{obs}} | L_i^R)^{m_i^R} dL_i^A dL_i^R. \quad (33)$$

This conditions on the missingness pattern and assumes it is ignorable for the model parameters. A mechanism depending on unobserved biology may need an additional missingness model.

Paired cell. Both likelihoods contribute. An ATAC update includes RNA-child feedback; an RNA update averages over uncertain ATAC parents.

RNA-only cell. Direct ATAC likelihood statistics satisfy $a^A = b^A = 0$. ATAC loadings remain latent parents of the observed RNA. Their updates use the prior and child terms in (11); there is no division by a zero precision and no substitution of a zero loading.

ATAC-only cell. The entire unobserved RNA branch integrates out:

$$\int p_\theta(L_i^R | L_i^A, X_i) dL_i^R = 1.$$

The cell informs the ATAC model but provides no direct evidence for the separately parameterized RNA conditional. RNA predictions are generated after fitting by averaging that conditional over inferred ATAC uncertainty.

11.2 Why an unobserved child should be collapsed

Retaining a factorized $q(L_i^A)q(L_i^R)$ for an ATAC-only cell adds

$$\mathbb{E}_q \log p_\theta(L_i^R | L_i^A, X_i) + H(q(L_i^R)) = -\mathbb{E}_{q(L_i^A)} \text{KL}\{q(L_i^R) \| p_\theta(L_i^R | L_i^A, X_i)\}. \quad (34)$$

This may penalize dependence solely because the variational family cannot represent it. It is a valid lower-bound gap, but there is no observed RNA evidence that warrants it. Integrating out the branch removes this avoidable gap. Equivalently, an unrestricted conditional variational factor could set $q(L_i^R | L_i^A) = p_\theta(L_i^R | L_i^A, X_i)$.

For each row, retain nodes with observed descendants and eliminate the remaining terminal branches in reverse generative order. In the implemented ATAC-to-RNA graph, RNA-only cells retain their ATAC parents; ATAC-only cells omit all RNA nodes from the fitting objective. Regularization for an omitted node is omitted for that row as well. Posterior prediction of eliminated nodes follows the generative ordering after the fit.

11.3 The same learning equations, two observation models

For node L_{ik}^m , compute a_{ik}, b_{ik} from modality m 's mask, precision, residual and feature moments. Then apply (14) with its active within- and between-modality children. Update F_k^m through the original ash normal-means solver using that modality's observed entries. Missing-modality predictions are never inserted as observations in a later feature or noise fit.

Removing cross-modality edges gives separate conditional-loading models. Removing *all* latent-covariate edges also gives the original scalar cEBMF normal-means updates. Preserving the original code path when no graph is required retains its initialization, options and finite optimizer behavior.

11.4 Identification and evaluation

Paired cells anchor the relationship between modalities. With no pairing, an unrestricted conditional distribution is generally not identified by the two marginal distributions; additional assumptions or anchors are needed. The paired subset should cover the states in which predictions will be used. Its mere existence is a software precondition, not a statistical sufficiency guarantee.

Evaluate denoising on paired cells separately from prediction of an entire missing modality. Hold out complete cell-modality rows to assess the latter. Fit preprocessing and choose hyperparameters without those held-out values. The Gaussian derivation applies to the toy observations and appropriately processed real measurements; a raw-count likelihood requires a separate extension of the normal-means calculation.

12 Positioning among learned priors and multi-omics models

The comparison concerns three separate choices: the generative loading prior, the observation model, and the posterior approximation. Changing one does not establish superiority in the other two.

12.1 Population priors and the relation to TwinEB

The closest comparison for flexible empirical-Bayes prior learning is TwinEB, introduced by Salehi, Nazaret, Shah and Blei [3]. TwinEB learns mixture priors on the entire row and column latent vectors and fits these priors jointly with a variational posterior. The paper develops Gaussian and Poisson matrix factorization. Its mixture-of-vectors construction must be distinguished from a product of independent scalar mixture priors: a shared component label can induce dependence among coordinates of a latent vector.

For the Gaussian formulation, write a diagonal-component row mixture as

$$p_{\text{pop}}(z) = \sum_{h=1}^H \omega_h \prod_{k=1}^K \varphi(z_k; m_{hk}, s_{hk}^2). \quad (35)$$

The scales in the paper are parameterized per component and coordinate (Appendix B). Its mean-field posterior approximation does not imply independence under the prior. For example, with $\bar{m} = \sum_h \omega_h m_h$,

$$\text{Cov}(z) = \sum_h \omega_h \text{diag}(s_h^2) + \sum_h \omega_h (m_h - \bar{m})(m_h - \bar{m})^\top.$$

The second term may have nonzero off-diagonal entries. Thus dependence between latent programs is not, by itself, a capability absent from TwinEB.

An exact relationship at the level of the loading prior. The vector-mixture form implies the autoregressive conditionals

$$p_{\text{pop}}(z_k \mid z_{<k}) = \sum_h \tilde{\omega}_{hk}(z_{<k}) \varphi(z_k; m_{hk}, s_{hk}^2), \quad (36)$$

$$\tilde{\omega}_{hk}(z_{<k}) = \frac{\omega_h \prod_{j<k} \varphi(z_j; m_{hj}, s_{hj}^2)}{\sum_r \omega_r \prod_{j<k} \varphi(z_j; m_{rj}, s_{rj}^2)}. \quad (37)$$

To prove the result, let $C_k(z_{\leq k}) = \sum_h \omega_h \prod_{j \leq k} \varphi(z_j; m_{hj}, s_{hj}^2)$, with $C_0 = 1$. The conditional in (36) equals C_k/C_{k-1} . Multiplying over k telescopes to C_K , which is (35).

Accordingly, a conditional Gaussian-mixture framework that permits the specific gates (37) contains this row-prior family exactly. An unrestricted conditional head can also vary the component means and variances with preceding coordinates, and incorporate observed covariates. However, an arbitrary finite ReLU network does not necessarily implement the log-quadratic Bayes gates exactly. A one-slab CGB is also a much smaller family than an unrestricted vector mixture. The result is a distributional inclusion under explicit assumptions, not a blanket inclusion claim for every registry configuration.

Any joint density admits an autoregressive factorization by the chain rule. Moreover, increasing the number of Gaussian mixture components already gives highly flexible continuous population models. The proposed ordering therefore does not establish that TwinEB cannot represent nonlinear dependence. The empirical question is whether the chosen conditional parameterization represents and learns scientifically relevant dependencies more efficiently at feasible sample

sizes and computational budgets. Exact atomic probabilities are an additional modeling choice; narrow continuous Gaussians approximate concentration near zero without assigning positive probability to zero itself.

Different structural assumptions. The present model explicitly separates modality-specific latent programs and specifies

$$p(A_i, R_i \mid X_i) = p(A_i \mid X_i)p(R_i \mid A_i, X_i)$$

through within- and between-modality conditional distributions. This allows external covariates to modify activation probabilities, amplitudes or scales at particular nodes. All such conditional factors belong to one normalized joint model; upstream updates include the downstream evidence described in (10).

TwinEB, as formulated in the cited paper, uses population priors for a single matrix. A multi-view extension could concatenate loading vectors and specify a suitable block observation model. That is a legitimate comparator construction, but it should be identified as an extension and implemented with the same missing-data likelihood and evaluation protocol rather than attributed to the published model.

The feature-factor assumptions differ as well. TwinEB’s column prior is a mixture on an entire feature’s latent vector. The principal implementation here uses independent ash priors for feature columns. Optional HMM priors instead connect adjacent feature positions within each factor. Neither construction is a strict superset of TwinEB’s vector column prior. Our Gaussian observation implementation also does not subsume the paper’s Poisson formulation.

Inference and the proposed contribution. Both approaches jointly learn priors and variational posterior approximations. Here the loading-coordinate posterior is profiled out, giving the local log-normalizer objective in (14). The no-latent-parent/no-child limit is the original cEBNM marginal-likelihood update. Numerical integration is needed for nonlinear child contributions, and a factorized posterior does not represent posterior covariance between loading coordinates.

The methodological candidate is a conditional-prior factorization framework with explicit activation probabilities, modality-specific programs, fixed side information and a cEBMF-compatible learning algorithm. Autoregressive decomposition and coordinate variational inference are established mathematics. Novelty and scientific value must rest on the model construction, useful inference and evidence at matched predictive and computational budgets.

Comparisons needed to support this positioning. Include TwinEB alongside independent cEBMF and the multi-omics comparators below. On a common Gaussian single-matrix benchmark, compare vector population mixtures and autoregressive mixtures at matched ranks and documented parameter and tuning budgets. For multi-view data, report any required extension of the competitor. Separately ablate the hierarchy, nonlinear means and scales, exact spikes, fixed covariates and optional HMM smoothing. Exponential slabs remain a separate extension. This distinguishes prior expressiveness, regularization, modality coupling and the inference algorithm.

Evaluation should include held-out observations, missing-modality prediction, predictive calibration, and computation at matched accuracy. For samplers, report effective sample sizes and convergence diagnostics rather than acceptance rates alone. For VI, assess the effect of its dependence assumptions against small reference problems. No result in this derivation establishes empirical superiority over TwinEB or the other methods.

12.2 Learned VAE priors and explicit program relationships

Learned latent priors are an established component of variational autoencoders. Their relevance here is both statistical and computational: they provide alternatives for representing dependence, but differ in the meaning assigned to latent coordinates and in the inference operations available under the observation model. Three distinctions are essential: prior flexibility does not imply an interpretable dependency graph; conditional independence within a mixture component does not imply marginal independence; and a tractable prior density does not imply a tractable posterior under an arbitrary decoder.

VampPrior: learned pseudo-inputs and shared mixture components. Tomczak and Welling [4] define the basic VampPrior by

$$p(z) = \frac{1}{H} \sum_{h=1}^H q_\phi(z \mid u_h), \quad (38)$$

where q_ϕ is the VAE encoder and the u_h are trainable vectors in the observation space (their equation 9). Pseudo-inputs are global model parameters optimized with the encoder and decoder. They need not be realistic observations, and they are not noisy copies of fitted loadings or sample-specific latent variables. With a diagonal Gaussian encoder, (38) is a vector Gaussian mixture whose parameters are tied through the encoder. It is not a normalizing flow.

Equivalently, introduce a sample-specific label $C_i \sim \text{Uniform}\{1, \dots, H\}$ and $z_i \mid C_i = h \sim q_\phi(z_i \mid u_h)$. The label is shared across coordinates of z_i , so the mixture can encode nonlinear and multimodal conditional dependence even when each component has diagonal covariance. Section 12.1 derives the corresponding coordinate conditionals. Neither diagonal components nor a diagonal variational encoder justify describing this population prior as independent. The latter restricts the approximate posterior for each observation, not the dependence of the mixture prior.

The same paper separately introduces a two-layer hierarchical VAE, with a VampPrior on its upper latent layer and a neural Gaussian conditional distribution for the lower layer (their equations 12–15). That extra continuous latent layer is not required by the basic VampPrior. In contrast, the present hierarchy specifies dependencies among the existing factorization loadings. Its mixture indicators are auxiliary labels for the specified component distributions, rather than a new continuous representation above the programs. Optional HMM states belong to a separate feature-sequence prior; the principal implementation here uses ash on the features.

Autoregressive-flow priors. The Variational Lossy Autoencoder of Chen et al. [5] parameterizes a latent prior by an invertible autoregressive transformation $z = f_\theta(\epsilon)$ of a simple continuous noise source. Its density follows from the change-of-variables identity,

$$\log p_\theta(z) = \log p_0(f_\theta^{-1}(z)) + \log \left| \det J_{f_\theta^{-1}}(z) \right|. \quad (39)$$

Such a prior explicitly introduces dependence between latent coordinates. A single affine Gaussian autoregressive step has Gaussian scalar conditionals in its ordering, but this restriction should not be generalized to arbitrary compositions and nonlinear flow families. Accordingly, complex distributional dependence itself is not a distinguishing capability of the present model.

The distinction is the specified semantics of its conditional heads: an exact activation probability, a chosen amplitude distribution, and an explicit parent set for each additive program. A smooth invertible flow with an absolutely continuous base distribution assigns no positive mass to an exact zero event. Exact spikes therefore require an additional discrete or mixed-support

construction; an ordinary continuous flow does not provide them by itself. Conversely, adding a neural conditional mixture does not automatically give more interpretable effects than a flow. Interpretation requires restrictions on the heads and stable definitions of the latent programs.

iVAE and conditional identifiability. Khemakhem et al. [6] study priors of the form

$$p(z \mid x) = \prod_k p_k(z_k \mid x), \quad (40)$$

with suitable exponential-family component distributions and an observed auxiliary variable x . Conditional factorization, together with additional assumptions on the decoder and variation of the conditional parameters, supports their identifiability results. It does not assert marginal independence after averaging over x . Their use of *exponential family* refers to the general statistical class, not specifically to an exponential slab distribution.

The present model allows direct dependence on latent parents even after conditioning on x . It therefore addresses a different modeling objective and does not automatically inherit iVAE’s identification guarantees. A directed loading prior is not a causal identification result, and coefficient interpretation remains conditional on the chosen ordering, scales and program definitions.

An interpretable linear-logistic specialization. Let $Z_{ik}^R = 1$ indicate the continuous slab of RNA loading R_{ik} ; when $Z_{ik}^R = 0$, set $R_{ik} = 0$. A restricted CGB prior can specify

$$\text{logit Pr}(Z_{ik}^R = 1 \mid A_i, R_{i,<k}, x_i) = \alpha_k + \beta_k^\top A_i + \eta_k^\top R_{i,<k} + \gamma_k^\top x_i. \quad (41)$$

For a parent-independent Gaussian slab, the coefficients govern activation probability rather than slab amplitude. Holding all other parents and context fixed, increasing A_{ij} by Δ multiplies the prior activation odds by $\exp(\beta_{kj}\Delta)$. This is a conditional association at a specified loading scale, not a causal effect or the posterior odds after observing both modalities. The latter also include the likelihood and child contributions derived earlier. Sign-constrained Gaussian means do not, by themselves, make loadings positive.

Equation (41) gives a more direct coefficient interpretation than a generic vector-mixture or flow parameterization. It does not prove superior empirical interpretability: latent rotations, permutations, scale changes and alternative program decompositions can change the apparent relationships. Anchoring or aligning programs and assessing stability across fits remain necessary for scientific use. The parent set consists of the prescribed preceding nodes. Unrestricted regressions of every loading on all the other loadings are not automatically compatible with a single joint distribution.

The current conditional CGB implementation uses ReLU neural gates. Equation (41) is a restricted model specification, not a coefficient interpretation of those fitted neural networks. Under fixed Gaussian slab parameters, the linear-logistic restriction also makes the continuous coordinate log-concave: a Gaussian log density plus Bernoulli log-likelihoods of affine predictors is concave. Neural gates and EMDN can subsequently be compared with this restriction when nonlinear activation, amplitude or variance relationships are scientifically required.

Observed context in a vector-mixture alternative. A conditional population mixture can incorporate the same fixed context:

$$C_i \mid x_i \sim \text{Categorical}(w_1(x_i), \dots, w_H(x_i)), \quad (42)$$

$$v_i \mid C_i = h, x_i \sim \mathcal{N}(\mu_h(x_i), D_h(x_i)), \quad v_i = (A_i^\top, R_i^\top)^\top. \quad (43)$$

Context may affect mixture probabilities, component means or covariance matrices; it need not become an uncertain latent variable. A conditional VampPrior could tie components through an encoder $q_\phi(v \mid u_h, x)$, whereas a directly parameterized mixture can learn these component functions without introducing that encoder. These are possible extensions, not a claim that the original VampPrior paper implements this multi-omics model.

Under diagonal component covariances, dependence between programs is mediated by the shared component label. Their conditional relationships can still be complex, but an effect of one program on another is usually distributed across the component parameters and responsibilities rather than summarized by a coefficient such as β_{kj} . Full component covariances permit additional dependence within each component.

Why Gaussian mixtures can simplify inference. Conditional on the feature factors, the observed row vector satisfies a linear Gaussian model $Y_i^{\text{obs}} = B_i v_i + \epsilon_i$, where B_i selects the observed entries from the two modality designs and $\epsilon_i \sim \mathcal{N}(0, \Sigma_i)$. For positive-definite Gaussian prior covariances in (43), the row conditional is a Gaussian mixture. Its component parameters are

$$S_{ih} = (D_h^{-1} + B_i^\top \Sigma_i^{-1} B_i)^{-1}, \quad (44)$$

$$m_{ih} = S_{ih}(D_h^{-1} \mu_h + B_i^\top \Sigma_i^{-1} Y_i^{\text{obs}}), \quad (45)$$

with weights proportional to

$$w_h(x_i) \mathcal{N}(Y_i^{\text{obs}}; B_i \mu_h, \Sigma_i + B_i D_h B_i^\top).$$

Here μ_h, D_h are evaluated at fixed x_i . These identities follow by completing the multivariate Gaussian square. They establish tractability of a row conditional at fixed factors and parameters, not an analytic posterior for the entire factorization. Missing entries or modalities are omitted from the observation vector and design. Feature updates must use moments consistent with the chosen posterior family; the block covariance correction is given in (31).

Both this mixture model and the direct hierarchy propagate information between modalities. The computational simplification is Gaussian conjugacy of the joint components; it does not arise merely from adding a latent label. In the direct logistic hierarchy, an upstream update instead contains nonlinear log-gate contributions from its children. Those contributions preserve the explicit model’s semantics while making its update nonconjugate. Exact sparsity patterns in vector mixtures require explicit lower-dimensional component supports, and exponential components do not generally preserve the unrestricted Gaussian block formulas above. Neither alternative is uniformly cheaper at comparable modeling accuracy.

Positioning and empirical consequences. For an interpretation-oriented study, the main model is a direct conditional loading hierarchy. Linear-logistic activation heads are a useful transparent comparator; the implemented neural gates do not have that coefficient interpretation. Its contribution must be assessed through interpretable conditional associations, useful predictions and joint uncertainty propagation, rather than through a claim that previous VAE priors cannot encode dependence. Conditional vector mixtures offer a relevant tractable comparator, and neural gates offer an expressiveness comparison within the same factorization framework. Differences in observation models, feature priors, ranks and tuning budgets must be controlled.

Finally, replacing the linear factorization by a nonlinear VAE decoder preserves the normalized conditional-prior model and its ELBO, but generally removes the exact quadratic normal-means likelihood. The analytic updates in this paper are therefore not automatically available for arbitrary VAEs. Learned priors and structured posterior approximations are related choices,

but neither posterior dependence nor exact spike probabilities follow from changing a prior alone.

12.3 Shared representations and modality-specific programs

MOFA+ uses a shared sample representation with modality-specific feature weights [7]; JIVE decomposes joint and individual variation [8]. The present construction instead specifies separate loading vectors and a conditional distribution between them. These are different structural assumptions, not an assertion that shared representations cannot encode the toy signal.

In the supplied example, two binary ATAC programs have four joint states. Four shared one-hot state indicators can represent the RNA programs, and each ATAC program is a sum of those indicators. A sufficiently rich shared linear representation therefore reconstructs both noiseless signals. The nonlinear relationship between a chosen pair of ATAC coordinates and four RNA coordinates does not prove a universal capacity limitation for MOFA or any other unrestricted shared-factor model. Comparisons must control rank, intercepts, parameter budgets and the task being evaluated.

Hierarchical nonlinear integration, as in CAVACHON [9], offers another way to organize modality relationships. MultiVI addresses paired and unpaired single-cell modalities with modality-appropriate generative models [10]; GLUE supplies a guidance graph linking features across modalities [11]. Their assumptions, decoders and information sources differ from this Gaussian conditional-loading construction. They are relevant empirical comparators, not special cases automatically covered by the present implementation.

Observed context and uncertain parents must also be distinguished. A fixed batch, donor or spatial covariate can enter a prior without becoming a latent loading. In contrast, substituting an estimated loading as a fixed covariate removes its uncertainty and downstream feedback. Neither joint fitting nor a directed prior by itself identifies causal regulatory effects.

12.4 Empirical comparisons required

At minimum compare independent cEBMF, fixed-loading covariates and the conditional-loading update with matched feature priors, ranks, initialization, noise treatment and tuning budgets. Assess several seeds and held-out cell-modality rows, including states sparsely represented in the paired subset. Separate numerical integration error from posterior approximation error; compare with exact enumeration or a well-diagnosed reference sampler on small problems. Assess conditional probability calibration and the stability of program definitions in addition to reconstruction error.

13 Implementation, example and validation

13.1 Two cEBMF objects, one joint fitting schedule

The maintained interface is

```
data = align_modalities(Y_atac, Y_rna, atac_ids, rna_ids)
settings = dict(prior_L="cgb", prior_F="norm",
                self_row_cov=True)
atac = cEBMF(data.atac, K=2, **settings)
rna = cEBMF(data.rna, K=4, **settings)
atac_fit, rna_fit = fit_joint(atac, rna, maxit=20)
```

The argument order defines the ATAC-to-RNA prior. Each model retains its own feature update and noise. The flag controls within-modality parents; the joint call supplies cross-modality parents. Passing `atac.L` as fixed RNA covariates is a different plug-in analysis. Repeated joint calls continue the same fit. Row IDs must be aligned before construction.

13.2 Numerical representation and supported scope

The implementation stores each coordinate's quadrature values, component weights, moments and continuous-and-label entropy. This state remains fixed until its next coordinate update. One uncertain parent is integrated with its full stored quadrature; multiple uncertain parents use fixed scrambled Sobol points. Child values and labels use component moments analytically. The target loading uses Gauss-Hermite integration around its Gaussian base components. These integration nodes do not define a discrete generative prior. The optimizer retains its best evaluated local profile objective; integration and finite training can still cause changes in the global approximate ELBO that are not monotone.

Conditional CGB effective slab variances are optimized in that objective. For sharp families, ω affects initialization; repeatedly multiplying variance estimates by ω is not treated as an exact variational M-step. The no-edge route calls the original scalar solvers directly, preserving their options and finite optimization behavior. Conditional graphs retain their initial rank and ordering. Graph pruning and posterior covariance between loading coordinates are not implemented in this path.

The former sampler is retained only as an explicitly selected experimental compatibility backend. It is not selected by `self_row_cov=True` alone. The primary example has no HMM, MCMC burn-in or frozen ash parameters.

13.3 Checks and their limits

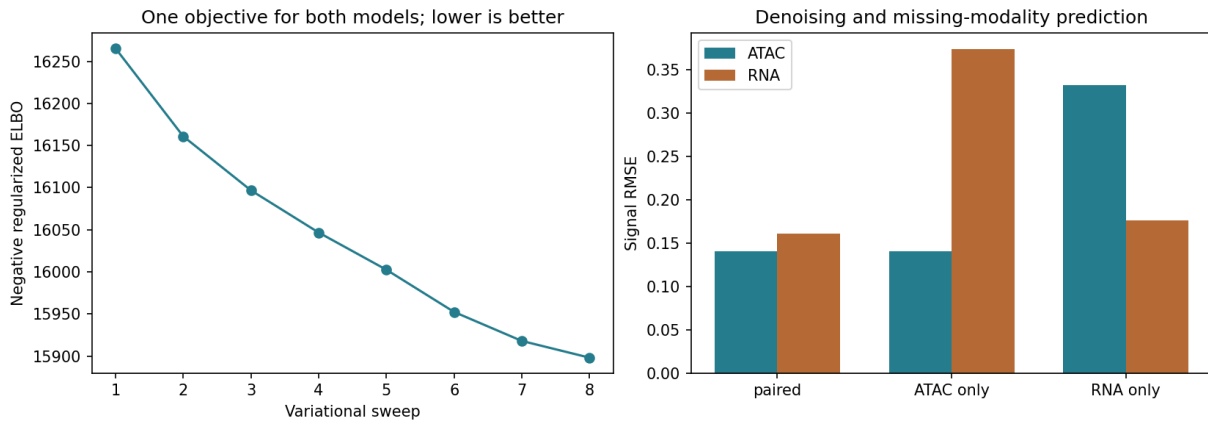
Regression tests compare mixture log normalizers, moments and entropy with analytic normal-means formulas, and compare a quadratic child tilt and its profile gradient with a closed-form Gaussian result. They check exact single-factor no-edge compatibility, ordinary fixed-covariate dispatch, noise updates, both matrix axes, collapsed missing children, inferred missing parents, unchanged stored variational coordinates and explicit option validation. Existing reference-sampler tests remain separate.

13.4 An executed toy illustration

The maintained notebook uses 120 cells, two ATAC factors, four RNA factors, 60 ATAC features and 80 RNA features. Eighty cells are paired; twenty have only ATAC and twenty have only RNA. Gaussian observation SD is 0.7. The fit uses CGB loadings, ash features, unit sparsity penalties, eight variational sweeps, five neural epochs per loading block, and a 12-unit network with one additional hidden layer. Numerical settings are 16 quadrature points per component and 16 parent-integration points. The random seed is 8.

Cell group	ATAC signal RMSE	RNA signal RMSE
Paired	0.141	0.161
ATAC only	0.141	0.374
RNA only	0.333	0.176

The errors use the noiseless simulation signal. RNA error in ATAC-only cells and ATAC error in RNA-only cells assess complete missing-modality prediction. No true loadings, factors or held-out measurements enter fitting or initialization. This single run demonstrates the workflow and its limitations; it does not establish improvement over an independent or plug-in comparator.



The shared numerical objective decreased in this run. That observation is not an unconditional convergence guarantee. Run the notebook from top to bottom and increase integration resolution before drawing performance conclusions. The revised short example completed in about 22 seconds in the tested CPU environment with one torch thread; this is not a cross-method timing benchmark.

14 Adversarial numerical review and device handling

Review added 16 September 2026. The target remains $p(L_k | L_{<k})$. Removing latent-covariate edges gives the same normal-means empirical Bayes subproblem; the implementation dispatches this case to the ordinary cEBMF solver. This algebraic statement does not assert that every finite neural optimization attains the subproblem optimum.

14.1 A correct formula can be an unstable implementation

Completing a Gaussian square using its natural parameters subtracts terms of order μ^2/s^2 . In float32 this failed for a prior with spike weight 0.2 and slab $\mathcal{N}(10, 10^{-8})$, with likelihood statistics $a = 2, b = 3$. The old numerical profile returned a posterior mean near 0.0125; the independent analytic posterior is approximately 1.59×10^{-29} . This is cancellation error, not Monte Carlo variability.

For each continuous component, compute instead

$$t = \mathbb{E}[s(P)^{-2}], \quad c = \frac{\mathbb{E}[\mu(P)s(P)^{-2}]}{t}, \quad (46)$$

$$h = \mathbb{E} \log \pi(P) - \frac{1}{2} \mathbb{E} \left[\log(2\pi s(P)^2) + \frac{(\mu(P) - c)^2}{s(P)^2} \right]. \quad (47)$$

Then the expected log prior is exactly

$$A(v) = h - \frac{1}{2}t(v - c)^2.$$

The spread in this expression is evaluated as an average of squared centered differences, never as a difference of large second moments. The height h retains the parameter-dependent normalization of $\exp(\mathbb{E} \log g)$; dropping it would change empirical Bayes learning.

Write $\alpha = a + t$, $r = t/\alpha$. The likelihood-completed Gaussian has mean $m = rc + b/\alpha$, variance $V = 1/\alpha$, and log mass

$$B = h + r(-ac^2/2 + bc) + \frac{b^2}{2\alpha} + \frac{1}{2} \log \frac{2\pi}{\alpha}. \quad (48)$$

Its tilted contribution is $\exp(B)\mathbb{E}_{N(m,V)} \exp D(v)$. For the zero atom the corresponding log mass before the child tilt is $\mathbb{E} \log \pi_0(P)$. These are algebraic rearrangements of the earlier derivation, not a different statistical model.

At a standard-normal quadrature node x_j , evaluate the slab log density as $-\frac{1}{2}\{\log(2\pi V) + x_j^2\}$. Combining it with $B + D(v) - \log Z$ gives the continuous-and-label log posterior used for entropy. This avoids another unstable natural-parameter polynomial. For child feedback use

$$\mathbb{E}[(L_c - \mu(P))^2 | Z_c = h] = \text{Var}(L_c | Z_c = h) + (\mathbb{E}[L_c | Z_c = h] - \mu(P))^2,$$

computing the within-component variance directly from centered nodes.

14.2 What was checked, and what was not established

Independent checks include exhaustive binary enumeration, analytic mixture posteriors (including zero likelihood precision), and a dense trapezoidal integral for a nonlinear logistic child tilt. Evidence, moments, entropy, and parameter gradients agree at the stated test tolerances. Direct child integration also checks the sparsity penalty and missing-branch mask. The ordinary

spike-normal posterior now uses log-domain component odds; normal-mixture likelihoods are no longer silently clipped in the tails.

These tests do not prove convergence of finite quadrature for every neural prior. Multiple-parent expectations use a finite fixed Sobol approximation; the stored quadrature representation also approximates each continuous coordinate. Local profile checkpoint selection is not a proof of global ELBO ascent under those approximations. The posterior is still factorized across loading coordinates. Better prior flexibility can lose predictive accuracy through optimization, approximation, or overfitting.

The saved tree benchmark predates these numerical fixes. Inspecting its checkpoints confirms float32 observations and loadings; the earlier claim of float64 fitting was incorrect. The narrow-slab bug could affect those fits, but attributing the performance gap to it requires matched reruns. Original source hashes and measurements are retained and have not been relabeled as results of the revised implementation.

14.3 CUDA contract and its validation limit

Loading moments, networks, quadrature states, optimizer state, residuals, and objective histories are kept on the selected device. Conditional checkpoint selection and mixture-EM convergence use device tensors. Observation indices and prediction tables are cached. Sobol tables are constructed once on CPU and transferred at graph setup. Ordinary CUDA neural solvers batch with device indices.

ASH’s automatically sized scale grid needs one host scalar for its length; its amplitudes stay on device. Supplying a fixed `scales` tensor to the ash prior removes that dynamic-size decision and still refits weights on every sweep. Fixed-grid choice is a modeling choice, not an automatically equivalent replacement for an adaptive grid. Rank pruning, boundary input validation, plotting, and explicit reporting may also synchronize.

The development environment has PyTorch 2.11.0+cpu and no CUDA runtime. CUDA-only tests cover complete warmed sweeps, partially paired modalities, optimizer-state residency, CPU/GPU numerical agreement, profiler transfers, and retained memory. They are skipped here, not reported as GPU passes. Hardware validation remains required before making a measured GPU speed or memory claim. The deprecated experimental sampler and optional L-BFGS route are outside the synchronization-free conditional/ash-EM contract.

References

- [1] W. Wang and M. Stephens (2021). Empirical Bayes Matrix Factorization. *JMLR* 22(120). <https://www.jmlr.org/papers/v22/20-589.html>.
- [2] W. R. P. Denault et al. (2025). Covariate-moderated Empirical Bayes Matrix Factorization. <https://arxiv.org/abs/2505.11639v2>.
- [3] S. Salehi, A. Nazaret, S. P. Shah and D. M. Blei (2024). Population Priors for Matrix Factorization. *TMLR*. <https://openreview.net/forum?id=AT9G5s1p0j>.
- [4] J. M. Tomczak and M. Welling (2018). VAE with a VampPrior. *AISTATS*, PMLR 84. <https://proceedings.mlr.press/v84/tomczak18a.html>.
- [5] X. Chen et al. (2017). Variational Lossy Autoencoder. <https://arxiv.org/abs/1611.02731>.
- [6] I. Khemakhem, D. P. Kingma, R. P. Monti and A. Hyvärinen (2020). Variational Autoencoders and Nonlinear ICA: A Unifying Framework. *AISTATS*, PMLR 108. <https://proceedings.mlr.press/v108/khemakhem20a.html>.

- [7] R. Argelaguet et al. (2020). MOFA+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome Biology* 21, 111. <https://doi.org/10.1186/s13059-020-02015-1>.
- [8] E. F. Lock et al. (2013). Joint and Individual Variation Explained (JIVE) for integrated analysis of multiple data types. *Annals of Applied Statistics* 7, 523–542. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3671601/>.
- [9] Hsieh et al. (2024). CAVACHON: a hierarchical variational autoencoder to integrate multi-modal single-cell data. <https://arxiv.org/abs/2405.18655>.
- [10] T. Ashuach et al. (2023). MultiVI: deep generative model for the integration of multimodal data. *Nature Methods*. <https://www.nature.com/articles/s41592-023-01909-9>.
- [11] Z. J. Cao and G. Gao (2022). Multi-omics single-cell data integration and regulatory inference with graph-linked embedding. *Nature Biotechnology*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9546775/>.